

Jenkins-Setup

Jenkins-CI/CD Setup:

Prerequisites:

- 1.AWS Ec2-server
- 2.Java
- 3.Jenkins

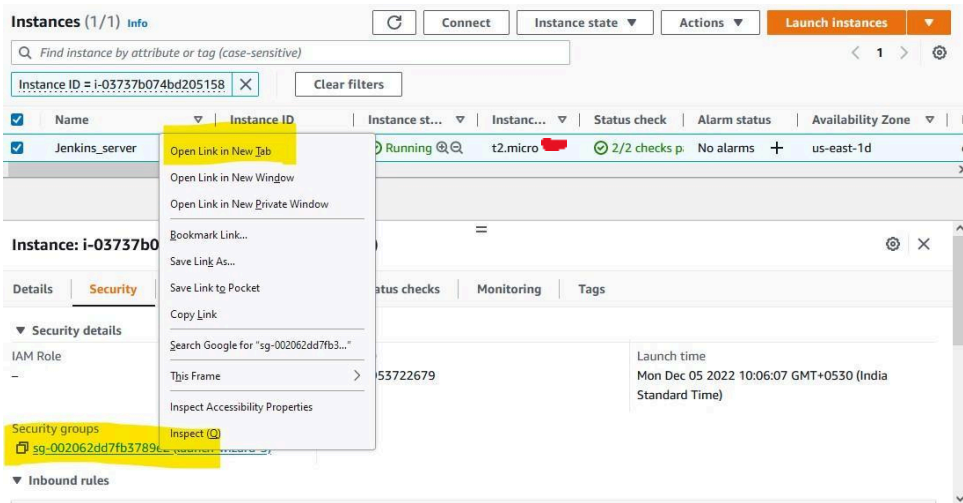
1.Launch New EC2 server in the AWS Cloud

The screenshot shows the AWS Management Console interface for the 'Instances' section. The instance 'Jenkins_server' (ID: i-03737b074bd205158) is listed with a status of 'Running'. The instance type is 't2.micro'. A red arrow points to this field, and the text 'Use t2.medium' is written above it. The instance details panel shows the instance is in the 'us-east-1d' availability zone with a public IPv4 address of 54.174.0.231.

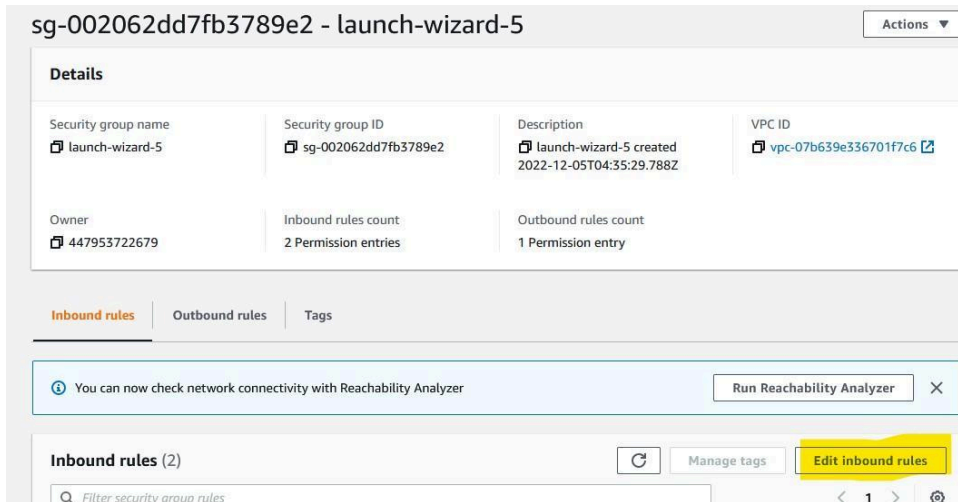
2.Once EC2 Created, Goto Security Group section:

The screenshot shows the AWS Management Console interface for the 'Instances' section, with the 'Security' tab selected. The instance 'Jenkins_server' (ID: i-03737b074bd205158) is listed with a status of 'Running'. The security group 'sg-002062dd7fb3789e2 (launch-wizard-5)' is listed under the 'Security groups' section.

3.Select Security Group, Open in new Tab:



4. Then select Edit Inbound Rules:



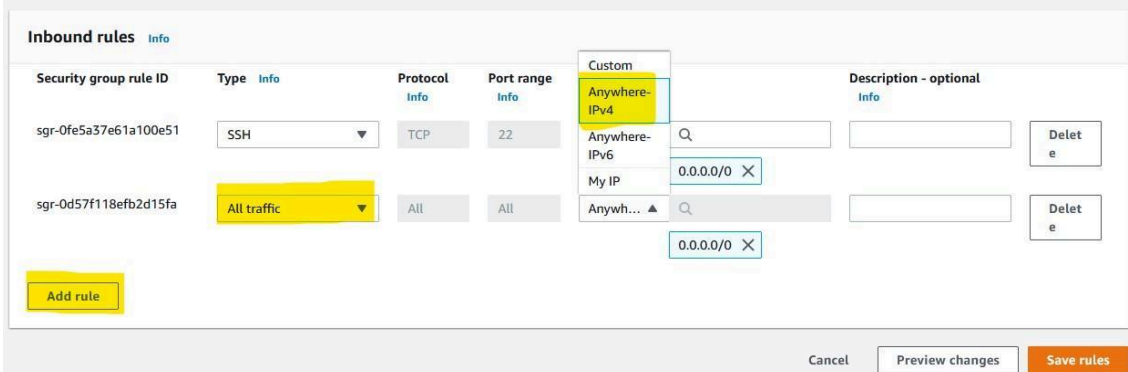
4. Click on Add rule

under Port Type select All Traffic and source as Anywhere-IPv4

EC2 > Security Groups > sg-002062dd7fb3789e2 - launch-wizard-5 > Edit inbound rules

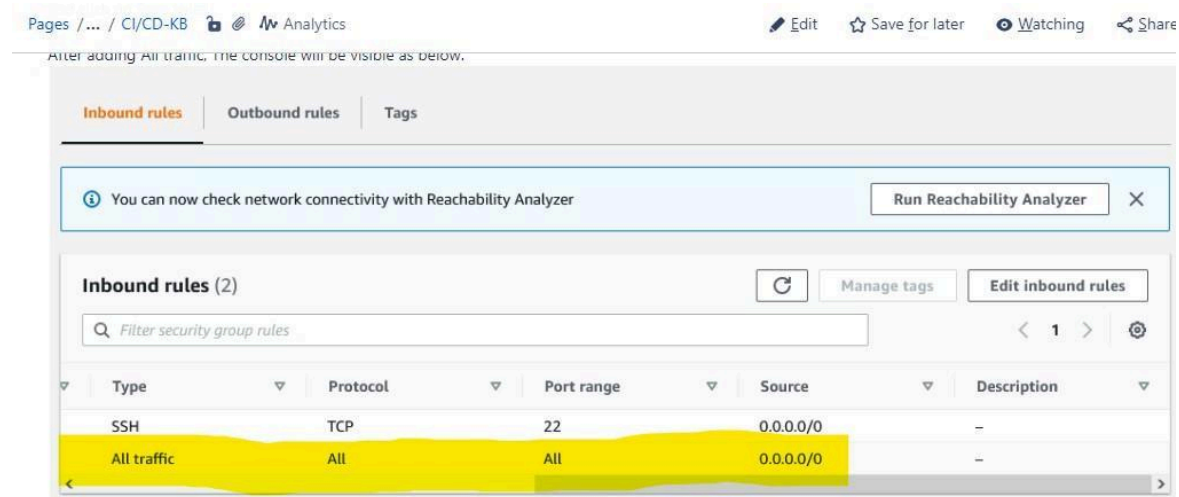
Edit inbound rules [Info](#)

Inbound rules control the incoming traffic that's allowed to reach the instance.



and click on Save rules

After adding All traffic, The console will be visible as below.



Step 1: SSH Into Your Instance

First things first, let's SSH into the Amazon Linux instance where we're setting up Jenkins. Open up your terminal and run:

```
ssh -i <your_key>.pem  
ec2-user@<your_amazon_linux_instance_ip>
```

This command lets you log in to your instance.

Just

replace `<your_amazon_linux_instance_ip>` with your instance's IP address and `<your_key>` with the actual key name.

Example: **ssh -i jenkins-server-key.pem
ec2-user@44.204.82.51**

All logged in? Perfect! Now, let's get down to business and start the Jenkins setup.

Step 2: Update the System

It's always a good practice to ensure that your system is up to date. Open a terminal window and enter the following command:

```
sudo dnf update
```

Step 3: Install Java

Jenkins is a Java application, so Java is a prerequisite. Install Java with the following command:

```
Download and Add the Amazon Corretto 21 Repo

sudo rpm --import
https://yum.corretto.aws/corretto.key

sudo curl -Lo /etc/yum.repos.d/corretto.repo
https://yum.corretto.aws/corretto.repo

sudo dnf install java-21-amazon-corretto -y
```

Verify the installation with:

```
java -version
```

Step 4: Add the Jenkins Repository

Next, add the Jenkins repository to your system with the following commands:

```
sudo wget -O /etc/yum.repos.d/jenkins.repo \  
https://pkg.jenkins.io/redhat-stable/jenkins.repo
```

Import a key file from Jenkins-CI to enable installation from the package:

```
sudo rpm --import  
https://pkg.jenkins.io/redhat-stable/jenkins.io-2  
023.key
```

Step 5: Install Jenkins

Now, install Jenkins by running:

```
sudo dnf install jenkins -y
```

Step 6: Start and Enable Jenkins

Enable and start the Jenkins service using:

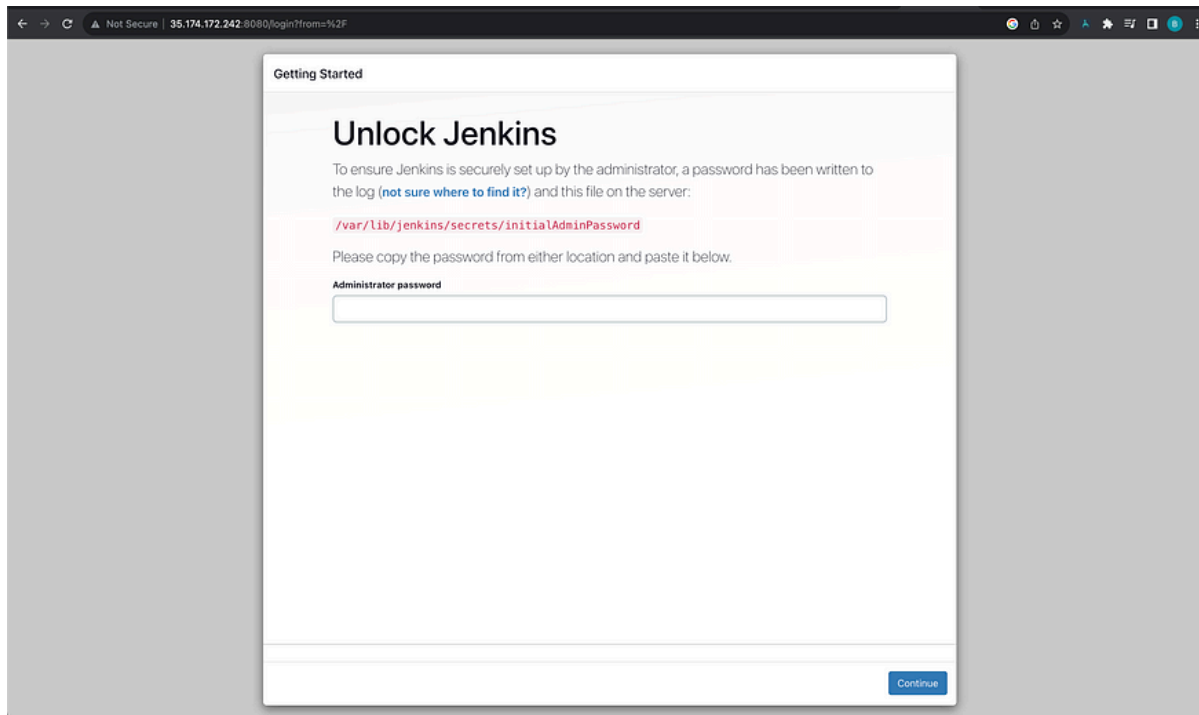
```
sudo systemctl enable jenkins  
sudo systemctl start jenkins  
sudo systemctl status jenkins
```

Step 7: Access Jenkins

Now, open your web browser and access Jenkins by navigating to:

```
http://your_amazon_linux_instance_ip:8080
```

You will see a setup wizard and be prompted to enter the administrator password.



##To get initial Admin password

```
sudo cat /var/lib/jenkins/secrets/initialAdminPassword
```

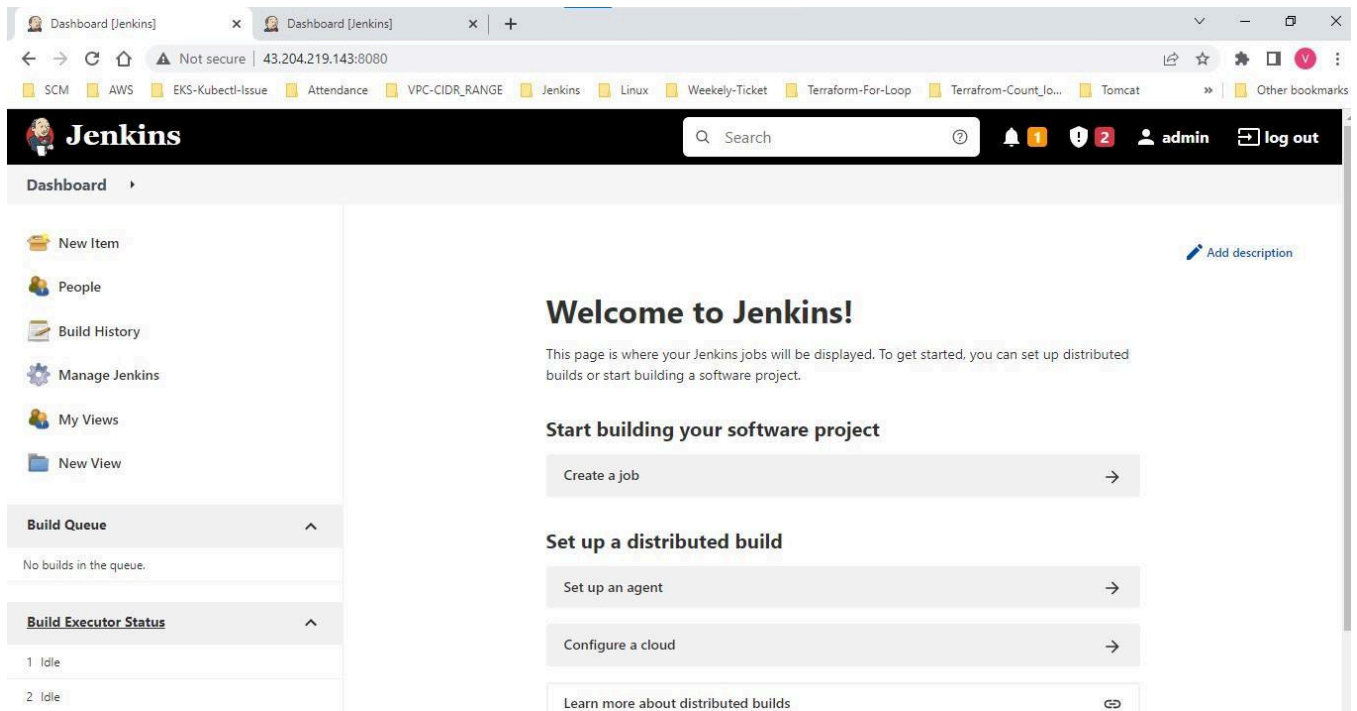


Signup with the username and passwd

username:admin

password:admin

email:admin@gmail.com



The screenshot shows the Jenkins Dashboard in a web browser. The browser's address bar displays "Not secure | 43.204.219.143:8080". The Jenkins logo is in the top left, and a search bar is in the top center. The top right shows a user profile for "admin" and a "log out" button. The left sidebar contains navigation links: "New Item", "People", "Build History", "Manage Jenkins", "My Views", and "New View". Below these are sections for "Build Queue" (showing "No builds in the queue.") and "Build Executor Status" (showing two idle executors). The main content area has a "Welcome to Jenkins!" message, a brief explanation of the dashboard's purpose, and a "Start building your software project" section with a "Create a job" button. Below that is a "Set up a distributed build" section with buttons for "Set up an agent", "Configure a cloud", and a link to "Learn more about distributed builds".

Dashboard [Jenkins] x Dashboard [Jenkins] x +

Not secure | 43.204.219.143:8080

SCM AWS EKS-Kubectl-Issue Attendance VPC-CIDR_RANGE Jenkins Linux Weekly-Ticket Terraform-For-Loop Terraform-Count_Jo... Tomcat » Other bookmarks

Jenkins Search ?

admin log out

Dashboard >

New Item

People

Build History

Manage Jenkins

My Views

New View

Build Queue ^

No builds in the queue.

Build Executor Status ^

1 Idle

2 Idle

Welcome to Jenkins!

This page is where your Jenkins jobs will be displayed. To get started, you can set up distributed builds or start building a software project.

Start building your software project

Create a job →

Set up a distributed build

Set up an agent →

Configure a cloud →

Learn more about distributed builds ↗